CONCRETE TECHNOLOGY

Subject Code : 23A01402T | Handwritten Study Notes

UNIT – II

FRESH CONCRETE

Syllabus covered in this unit

Steps in the manufacture of concrete – proportioning, mixing, placing, compaction, finishing, curing and their types – properties of fresh concrete – workability – factors – tests – setting times – effect of time & temperature – segregation & bleeding – mixing & vibration – ready mixed concrete – shotcrete

1. INTRODUCTION

- **Fresh (green) concrete** = concrete from the time of mixing till it sets (about 2–4 h). Good fresh concrete must be **mixed, transported, placed, compacted and finished easily without segregation**.
- Quality of hardened concrete (strength, durability) depends more on **how concrete is made & handled** than on mix design alone.



Stages in the manufacture of concrete (each stage affects final quality)

Concrete = Cement + Water + Fine aggregate + Coarse aggregate (+ admixtures)★

Typical volume : cement 7–15 %, water 14–21 %, air 1–6 %, aggregates 60–75 %. Normal concrete density ≈ 2400 kg/m^3 (plain) / 2500 kg/m^3 (RCC).

2. STEP 1 – PROPORTIONING (BATCHING)

Selecting right quantities of cement, FA, CA, water & admixtures (mix design – Unit V).

- **Volume batching** : quantities by volume using gauge boxes (farma) of size 35 × 30 × 25 cm = 1 bag cement (0.035 m^3). Used for small works, nominal mixes (M5–M20). Needs **bulking correction** for sand.
- **Weigh batching** : by mass using weigh-batchers (spring, beam, electronic load cells) – **accurate**, recommended for M25 and above (IS 456 cl. 9.2.2). Tolerance : cement, water ±2 %,

aggregates ±3 %, admixtures ±3 %.

- Water measured by volume (graduated tank / flow meter) ; admixtures by weight or volume.
- Types of batching plants : manual, semi-automatic, fully automatic, computerised (RMC plants).

Nominal mixes (IS 456, Table 9)

Grade	Mix ratio (C:S:A)	Total water per 50 kg cement (L max)	Use
M5	1 : 5 : 10	60	PCC / levelling
M7.5	1 : 4 : 8	45	PCC foundations
M10	1 : 3 : 6	34	PCC, footings
M15	1 : 2 : 4	32	Plain RCC works
M20	1 : 1.5 : 3	30	General RCC

3. STEP 2 – MIXING

Aim : **uniform mix – every aggregate particle coated with cement paste.**

3.1 Hand mixing

- On a clean, water-tight platform. Sand + cement mixed dry till uniform colour, then aggregate, then water added in a **depression** gradually. Turned 3–4 times. Used only for small jobs; **add 10 % extra cement**.

3.2 Machine mixing (IS 1791 batch type mixers)

Type	Working	Use
Tilting drum	Drum axis tilts to discharge	Small–medium jobs, dry/stiff mixes
Non-tilting drum	Chute inserted for discharge	Medium jobs
Reversing drum	Blades inside, drum rotation reversed to discharge	Large works, good mixing
Pan (forced action)	Rotating blades in a fixed pan	Stiff / lean / light-weight / precast
Dual-drum, twin-shaft	Continuous / high output	RMC plants, dams

- **Order of feeding** : ¼ water + coarse aggregate → cement + sand → remaining water ; admixture along with last water.

- **Mixing time** : not less than **2 min** after all materials are in the drum (IS 456 cl. 10.3.1), at **14–20 rpm** ; over-mixing → water loss, segregation, grinding of aggregate, air loss.
- Mixer should be cleaned before each fresh batch of different mix; first batch should have 10 % extra cement.
- **Mixing water** per bag of cement must be accurate – excess water ↓ strength.

4. STEP 3 – TRANSPORTING

- Concrete should be moved **as quickly as possible** without segregation, loss of slump or contamination (initial set ~ 30 min).
- Methods : **wheel-barrows / buggies** (short, level), **manual carriers**, **dumpers / trucks**, **belt conveyors** (≤ 20°, dry mix), **chutes** (≤ 1V : 2.5H), **skip & hoist / crane & bucket** (high-rise), **transit mixers** (RMC), **concrete pumps** (horizontal up to 300 m, vertical up to 100 m +), **pneumatic placers** (tunnels).
- Pumpable concrete : slump 80–150 mm, ≥ 300 kg/m^3 cementitious, rounded aggregates, MSA ≤ ⅓ pipe dia, well-graded.
- Precautions : water-tight clean containers, avoid drop > 1.5 m, no re-tempering.

5. STEP 4 – PLACING

- Prepare : clean formwork, remove loose material, apply oil/release agent, wet the subgrade and old concrete, fix reinforcement with cover blocks, **check levels & alignment**.
- Place concrete **as near as possible to final position**, in **layers 150–300 mm** thick, each layer compacted before the previous one sets (avoid cold joints).
- **Free fall ≤ 1.5 m** (use tremie, chute, elephant trunk) ; do not use vibrators to move concrete laterally; start from the far end of formwork.
- **Underwater concreting** : tremie method, Hydrophobic pumps, bags, **prepakt (grouted aggregate)**; do not allow water flow – cement washout.
- **Pumping / Shotcrete / Preplaced aggregate** are special placing techniques.
- **Construction joints** : location at points of minimum shear (≈ ⅓ span), roughened & cleaned, cement slurry/bonding agent before next lift ; **Expansion joints** (gap 20–25 mm for lengths > 45 m) & **Contraction joints**.

6. STEP 5 – COMPACTION

Purpose : expel entrapped air (5–20 % in fresh) , make dense, avoid **honeycombing** . **1 % voids ↓ strength by 5–6 %**; 5 % voids ↓ strength 30 %.

6.1 Hand compaction

- **Rodding** (16 mm bar, pocking), **tamping** (floors, slabs), **ramming** (dry earth-moist concrete in foundations) – only for workable mixes (slump > 100 mm, upto 150 mm per layer).

6.2 Mechanical compaction – vibrators

Needle / immersion (internal) vibrator	Form / shutter (external)	Surface vibrator (screed, plate)	Table vibrator
Most common, 25–75 mm dia 3000–12000 rpm, RC members	Fixed on formwork thin, congested, precast	Slabs, roads, pavements depth ≤ 150 mm	Precast units / lab cubes

- **Immersion vibrator** : insert vertically, quickly, withdraw slowly (≈ 3 cm/s); **radius of action ≈ 6–10 × needle dia (≈ 0.3–0.6 m)** ; spacing ≈ 1.5 × radius; time 5–15 s per point until air bubbles stop & paste glistens; penetrate 100 mm into previous layer; keep ≥ 75 mm from forms (avoid bar vibrating).
- **Over-vibration** → segregation, bleeding, laitance, **honeycombing** if under-vibrated.
- Frequency : 3000–12 000 vibrations/min ; amplitude 0.5–2.0 mm. Resonance with grains for high frequency.
- **Other methods** : **centrifugation / spinning** (pipes, poles), **pressure/ vacuum dewatering** , **jolting / shock tables** , **self-compacting concrete** (no compaction).

7. STEP 6 – FINISHING

- Operations on the surface of slabs / floors to produce a **smooth, level, durable surface** .
- **Screeding** : strike-off with straight edge to the level. **Floating** (wood / magnesium float) after bleed water disappears – seals surface. **Trowelling** (steel trowel) – smooth, dense surface. **Brooming** : skid-resistant finish for pavements.
- **Power floating** (large floors), **vacuum dewatering** (removal of 15–20 % water from slab : ↑ early strength, abrasion), **dry shake hardeners** , **exposed aggregate** , **stamped / coloured** finishes.
- **Never sprinkle dry cement** on surface – crazing. Do not finish while bleed water is on

surface.

- Formed (vertical) surfaces : rubbing, bush-hammering, sand-blasting, plastering (not recommended).

8. STEP 7 – CURING

Curing = maintaining satisfactory **moisture & temperature** in concrete in first days after placement so cement hydrates fully. Without curing, strength may be only 50 – 60 % of the potential and surface becomes porous.

- IS 456 cl. 13.5 : keep moist **≥ 7 days** for OPC; **≥ 10 days** with mineral admixtures / PPC ; **≥ 14 days** in hot dry weather (or for ordinary Portland with pozzolana & ...). Start curing after the final set (within 24 h), formwork loosening not earlier than prescribed stripping times.

Methods of curing

Method	Description	Use
Ponding / immersion	Dykes of mortar, water ponded on surface	Slabs, floors, roofs
Spraying / fogging	Fine mist or sprinkling; continuous	Vertical walls, columns
Wet covering	Wet hessian, gunny bags, sand, saw dust, straw	Columns, steps
Membrane (curing compound)	Spray resin / wax / PVA / bitumen emulsion forms water-tight film; white pigmented reflects heat	Pavements, where water scarce, large areas
Polythene / plastic sheet	Impermeable sheet seals moisture	Slabs, precast
Steam curing	Low-pressure steam 40–80 °C, 12–18 h at atmos. pressure; high pressure **autoclave** 8–12 atm	Precast, factory elements, ↑ 28-d strength in 1 day
Hot water / infra-red / electrical	Accelerated curing (IS 9013) for quick strength test of cubes	Precast, test

- Effect of hot/cold weather : concreting in **hot weather (>40°C)** – retarders, cool water, early morning ; **cold weather (<5°C)** – hot water, covers, accelerators, heating; must not allow freezing before 3.5 MPa.

9. FRESH CONCRETE – WORKABILITY

Workability (IS 6461 Part 7) ★

"Property of fresh concrete which determines the **ease and homogeneity** with which it can be **mixed, placed, compacted and finished**." Broad components : **consistency / mobility** (flow), **compactability** (ease of removing air), **stability** (resistance to segregation & bleeding / cohesiveness).

- Workability is a **combined property** – no single test measures it fully.
- **Useful internal work** (energy for full compaction) + **internal friction** between particles (and with forms/reinforcement).
- Required workability depends on **member (thin, congested rebars), compaction method, placing method.**

Placing condition (IS 456 cl. 7.1)	Degree	Slump (mm)
Blinding concrete, shallow sections, **pavers**, mass concrete / **vibrated**	Very low	See CF 0.75–0.85
Mass concrete; lightly reinforced slabs, beams, walls, columns (vibrated)	Low	25 – 75
Lightly reinforced sections without vibration; heavily reinforced (vibrated)	Medium	50 – 100 / 75–100
Trench fill, in-situ piling, heavily reinforced w/o vibration	High	100 – 150
Tremie concrete, pumped	Very high	150 – 180 + (flow test)

10. FACTORS AFFECTING WORKABILITY

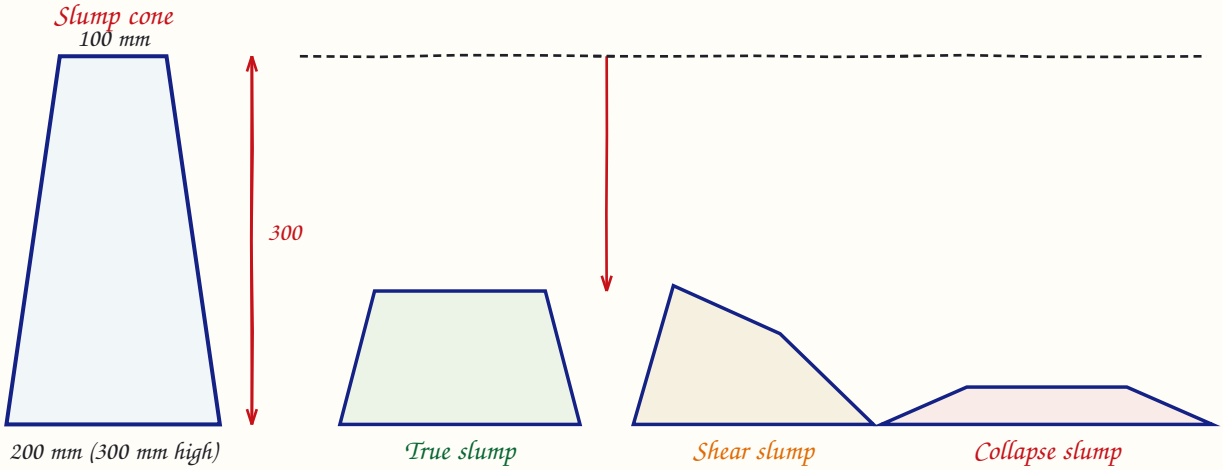
1. **Water content** (most important) : more water → higher slump; each 1 % ↑ water ≈ 6–10 mm slump; **excess water → bleeding, segregation, ↓ strength**.
2. **Cement paste** (quantity & consistency) : more paste → more lubricating → workable; very fine cement needs more water.
3. **Aggregate/cement ratio** : lean mix (more aggregate) is harsh & less workable.
4. **Aggregate size** : larger → less surface → more workable. **Shape** : rounded ≫ angular > flaky. **Texture** : smooth > rough. **Porosity** : porous aggregate absorbs water.
5. **Grading** : well-graded → fewer voids → more workable; sand content (FA %) ↑ → cohesive but less workable beyond optimum.
6. **Use of admixtures** : plasticizers, superplasticizers, air entrainers ↑ workability ; fly ash ↑ ;

silica fume ↓.

7. **Time & temperature** (slump loss), **humidity**, **wind**; **mixing time**; **cement type & fineness** (RHC less workable); **method of compaction**.

11. MEASUREMENT OF WORKABILITY

11.1 Slump test (IS 1199 : 1959) – consistency

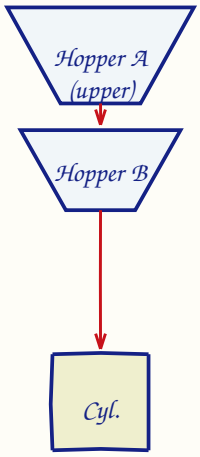


- Apparatus : frustum cone **100 mm top dia, 200 mm base dia, 300 mm high**, tamping rod 16 mm dia, 0.6 m long, bullet-nosed ; base plate, scale.
- Procedure : cone on non-absorbent plate, damp → fill in **3 layers** (each 1/3 height) → each tamped **25 times** → strike off → lift cone vertically in 5–10 s → measure subsidence (to nearest 5 mm) from top of cone.
- **True slump** : uniform subsidence – valid ; **shear slump** : one half slips – repeat test, indicates harsh/non-cohesive mix ; **collapse** : too wet / high water (too high a slump – not used for > 175 mm).
- Merits : simple, cheap, portable, used at site. Demerits : not suitable for very dry (< 10 mm) or very wet; does not measure compactability ; **does not differentiate between mixes of same slump with different aggregate.**

Slump values (typical) ★

Very low 0–25 mm : roads (vibrated) · Low 25–75 : foundations, mass concrete · Medium 50–100 : slabs, beams, columns · High 100–150 : heavily reinforced sections · Flowing > 150 mm with SP.

11.2 Compaction factor test (Glanville) – compactability

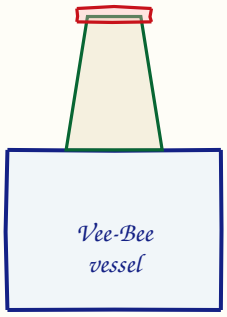


Compaction factor (Glanville)
= wt. partially compacted / wt. fully compacted
Trap doors released → concrete falls A → B → cylinder
CF 0.78 very low · 0.85 low · 0.92 medium · 0.95 high

- Suitable for **low workability** (slump 10–50 mm) – CF range **0.7 – 0.95** ; more sensitive than slump. Not suitable for CF > 0.92 (very wet) as concrete sticks in hoppers.
- Dimensions : upper hopper 25 cm top dia, 12.5 cm bottom, 22.5 cm high; lower hopper 22.5 cm dia, 22.5 cm high; cylinder 15 cm dia × 30 cm.
- CF = W_1 / W_2 ; W_1 = weight of partially compacted concrete in cylinder (falling only by gravity), W_2 = weight of fully compacted concrete (same cylinder, vibrated / rodded).

11.3 Vee-Bee consistometer test (Bahrner) – very dry mixes

Vee-Bee consistometer



Procedure :
1. Slump cone placed in cylindrical vessel on vibrating table
2. Fill in 3 layers, remove cone, set transparent disc on top
3. Start vibrator + stop-watch
4. Time (s) until disc face is fully covered with mortar
= **Vee-Bee degree** (Vebe seconds)
Suits very dry mixes : 3–30 s; slump test unsuitable.

11.4 Flow table test (IS 9103 / 1199)

- Table 76 cm square, cone 25 cm base × 17 cm top × 12 cm high placed on it ; after lifting the cone **table lifted & dropped 12.5 mm, 15 times in 15 s**. Flow % = (D – 25)/25 × 100 where D = mean spread diameter in cm. **Flow 0–150 % (medium 40–60 %)**. Good for **high workability / flowing mixes**; also detects segregation.

11.5 Other tests

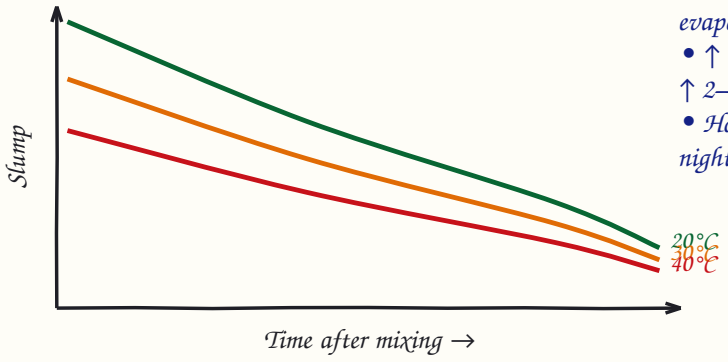
- **Kelly ball test** (ASTM C360) : 15 cm dia, 13.6 kg hemispherical ball dropped on surface; penetration depth ≈ ½ slump (2.5 cm penetration = 5 cm slump) ; fast, field test ; min depth 20 cm required.
- **Nasser's K-tester** (probe), **Remoulding test (Powers)**, **Penetration test**, **Ball penetration** .

Degree of workability	Slump (mm)	Compaction factor	Vee-Bee (s)
Very low	–	0.78	20 – 12 (≥ 20 s)
Low	25 – 75	0.85	12 – 6
Medium	50 – 100	0.92	6 – 3
High	100 – 150	0.95	3 – 0

12. SETTING TIME OF CONCRETE (IS 8142)

- Not same as cement setting time ; found by **penetration resistance test** on **mortar sieved through 4.75 mm sieve** from fresh concrete.
- Needle of different bearing area is pressed (≈ 25 mm) and **penetration resistance = force / area** vs elapsed time plotted:
  - **Initial set** : resistance = **3.5 N/mm²** (500 psi) ; **Final set** : **27.6 N/mm²** (4000 psi).
- Typical : initial 3–4 h, final 5–8 h ; retarders delay, accelerators / high temp reduce. It governs finishing, vibration re-start (stop before initial set) and formwork removal.

13. EFFECT OF TIME & TEMPERATURE



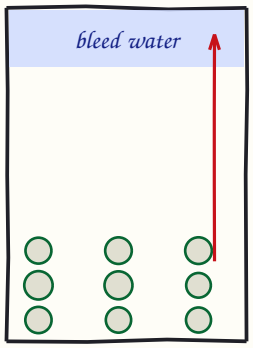
- Slump falls with time (loss of water by evaporation, absorption & hydration)
- ↑ 10 °C → slump ↓ ≈ 25 mm, water demand ↑ 2–3 %, setting time ↓ by ~ 25 %
- Hot weather : retarders, chilled water / ice, night concreting, shading, wet aggregates

- **Slump loss** : ≈ 25 – 50 mm per hour typically ; loss faster with PCC & $CaCl_2$. Hence **retempering** (adding water) not allowed (unless within limits, with SP addition).
- Remedy : add super-plasticiser at site (re-dosing), retarding plasticisers, cool materials, use

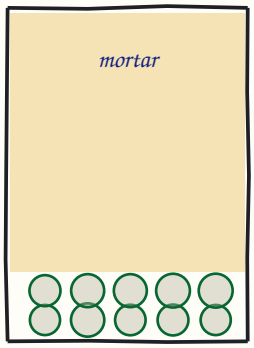
shade.

- In **cold weather** : slow hydration, set delayed ; protect against freezing & high mixing temperature (≥ 10 °C).

14. SEGREGATION AND BLEEDING



Bleeding



Segregation

Bleeding → water rises
laitance, weak top, voids
under bars/aggregate
Segregation → coarse grains
sink, mortar separates
honeycomb, weak, porous

14.1 Segregation

- **Separation of coarse aggregate from mortar** (or water from paste), making concrete **non-uniform** .
- Types : (a) coarse particles settle / separate from mix (dry mixes) ; (b) paste (cement-water) separates from aggregate (wet mixes) ; (c) water separates (bleeding).
- Causes : **too wet or too dry mix**, poorly graded / gap-graded aggregate, high MSA, **high fall (>1.5 m)**, over-vibration, rough handling, chutes with steep slope, long conveyor drop.
- Effects : honeycombing, weak spots, porosity, poor bond, erratic strength.
- Remedies : **good grading, adequate fines, lower w/c, air entrainment, limit fall height, proper vibration, SP + VMA**.

14.2 Bleeding

- **Water rising to the surface** after compaction, because solids settle (cement, 3.15 : water 1) and water is squeezed out; rate 0.1–0.5 % ; **bleeding capacity** 2–4 % of mix water.
- Causes : excess water, **low fines**, coarse cement (low specific surface), low temperature, over-vibration, deep sections, lean mixes.
- Effects : **laitance** (weak scum layer on top), porous, weak top surface (poor abrasion), poor bond with reinforcing bars & aggregates (pockets of water below them), **plastic settlement cracks**, reduces strength ≈ 10 – 15 %; sometimes beneficial – prevents plastic shrinkage by

replenishing surface water.

- Control : **finer cement** (more C_3A, alkalis), reduce w/c, ↑ fines (filler / fly ash), **air entrainment**, adequate mixing, re-vibration at proper time (**revibration**).
- Test (IS 1199 / ASTM C232) : measure volume of bleed water in 10 L container.

Difference ★

Segregation = separation of **aggregates from the mortar**; Bleeding = separation of **water from the solids** (a special form of segregation).

15. MIXING AND VIBRATION – IMPORTANT POINTS

- **Mixing** : IS 456 → mix for ≥ 2 min ; mixers must have revolution counter, water tank ; **initial batch** with extra cement ; do not use re-mixed concrete after initial setting ; time from mixing to placing **≤ 30 min** (mix) and ≤ initial set.
- **Vibration** : immersion type 25–75 mm needle dia, frequency 12 000 rpm ; avoid contact with reinforcement ; vibrate ≤ 20 s at a point ; **revibration** within permissible time ↑ strength 10–15 % & bond.
- **Needle spacing** ≈ 1.5 × radius of action ; formwork must be water-tight.
- **Safe stripping times (IS 456 cl. 11.3)** : vertical sides 16–24 h ; slab soffits (props left) 3 d ; beam soffit 7 d ; slab props 7 d (≤ 4.5 m span) ; slab soffit with props 14 d ; beams 14 d (spans > 6 m: 21 d).

16. READY MIXED CONCRETE (RMC) – IS 4926

- Concrete **produced in a central batching plant** (computerised) & delivered in **transit mixers (agitators)** to site in plastic state.

Type	Description
Central-mixed	Fully mixed at plant, hauled in agitator trucks (slow agitation) or non-agitating trucks
Transit-mixed (truck-mixed)	Dry batched at plant, mixed in truck drum enroute (70–100 revolutions at mixing speed 6–18 rpm)
Shrink-mixed	Partially mixed in plant mixer, mixing completed in truck

- Advantages : **better quality control**, speed, economy at large volume, less labour, less storage space & dust, wastage, **high strength (SP, SF) achievable**, environment.
- Limitations : transport delays/traffic ; slump loss (use retarders/SP) ; cost for small quantity;

need for pump & trained labour; plant location.

- Delivery : discharge within **2 h** of adding water ; max haul time depends on temp. Drum capacity 4–9 m^3. Tests : slump & cube strength at delivery point.

17. SHOTCRETE (GUNITE) – IS 9012

- Mortar / concrete **pneumatically projected at high velocity** (90–120 m/s) onto a surface – **self-compacting by impact**, no formwork.
- **Dry-mix process** : dry cement + sand/ aggregate (≤ 10 mm) conveyed by compressed air (0.3–0.5 MPa) through hose ; **water added at the nozzle** ; Rebound loss 15 – 50 %.
- **Wet-mix process** : ready-mixed wet concrete pumped to nozzle, air added at nozzle for projection ; lower rebound (5–15 %), more output, better quality.
- Accelerators (aluminate, silicates) for quick set on walls / overhead ; steel mesh / fibres for reinforcement ; **Rebound** = material bouncing off — avoid trapping in new layers.
- Properties : high early strength (≥ 20 MPa in 28 d, up to 50 MPa), low w/c (0.35–0.45), dense, good bond.
- Applications : **tunnel linings, slope stabilisation, canals, swimming pools, repair of concrete structures, domes, shells, rock support.**

Quick revision – Unit II ★

- Order : batching → mixing (≥ 2 min) → transport → placing (fall ≤ 1.5 m) → compaction (vibrator) → finishing → curing (7 / 10 days).
- Tests : slump (3 layers × 25), CF = W_1/W_2 (0.78–0.95), Vee-Bee seconds, flow table; setting 3.5 & 27.6 N/mm^2 (IS 8142).
- Segregation (aggregate from mortar) ≠ bleeding (water rises → laitance).
- RMC (central, transit, shrink) ; shotcrete (dry & wet mix).